Indian Institute of Technology Madras

CS6886: Systems Engineering for Deep Learning

Project Report

# Exploring LLS: Granularity, Early Exit and Scaling LLS

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Code: `https://github.com/Yukash09/CS6886_Project`

**Abstract**

Local Learning Synchronization (LLS) [1] trains each layer with its own local loss against a fixed periodic basis, removing end-to-end backpropagation (BP) and most of its training memory at a cost in accuracy. We explore two of its design axes. For *granularity*, we group layers into blocks that share one LLS head and find that where the cuts go matters more than how many heads remain: merging the early layers closes 89% of the LLS–BP accuracy gap on VGG8 at 1.29× the per-layer LLS memory, and 94% on MobileNetV1 at 2.56× (BP needs 8.21×), while simple uniform or pooling-aligned rules do not transfer between models. For *early exit*, we use the LLS heads as free exits: within one point of the full network, MobileNetV1 runs at 10–15% of its compute and VGG8 at 57–68%, a gap set by the architecture rather than the stopping rule. Coarser partitions give better exits, fixed heads match learned linear exits except where the head input is very wide, and BP with jointly trained exits remains better at every cost. The per-layer heads are also interpretable: they are calibrated, they show where a network stops improving, and their agreement tracks how hard an input is.

## 1 Introduction

Training a network with backpropagation (BP) requires storing the activations of every layer until the backward pass, so training memory grows with depth and limits on-device learning. Local learning rules avoid this by training each layer with its own loss, and LLS [1] is a recent rule that does so with a fixed, parameter-free target per layer. The price is accuracy: per-layer LLS trails BP by several points. In this project we study how far that gap can be closed and what else the local heads can be used for. Specifically, we ask three questions:

1. **Granularity:** instead of one LLS head per layer, can a block of layers share one head, and which grouping gives the best accuracy for a given training memory?
2. **Early exit:** since every LLS head already produces a class prediction, can these predictions be used to stop inference early, and do they tell us anything interpretable about the network?
3. **Scalability:** does LLS scale to larger models such as ResNet?

## 2 Prior Work

Local learning rules replace the back-propagated error with a signal that each layer can compute on its own. Feedback alignment and its variants send the error or the label through fixed random matrices [3, 4, 5], Forward-Forward replaces the backward pass with a second forward pass [6], and local-classifier methods attach a small trainable classifier to each block [7, 8]. The last family scales best to deep networks, but each classifier adds parameters and compute.

LLS [1] keeps the idea of a local classifier but makes it fixed. Each layer $l$, with (pooled) output $h^{(l)} \in \mathbb{R}^{T_l}$, is assigned a basis $B^{(l)} \in \mathbb{R}^{C \times T_l}$ whose row $c$ is a square wave with a class-specific frequency $f_c$. The layer is trained to make its output resemble the wave of the correct class:

$$\mathcal{L}^{(l)} = \mathrm{CE}\left(h^{(l)} B^{(l)\top},\, y\right), \qquad \delta^{(l)} = (p^{(l)} - y) B^{(l)}, \tag{1}$$

where $p^{(l)}$ is the softmax of the projection. Because the learning signal $\delta^{(l)}$ depends only on the layer's own output, no error crosses layers and only one layer's activations need to be kept at a time. We refer to the projection $h^{(l)} B^{(l)\top}$ as the *LLS head* of layer $l$. Early-exit networks [10, 11], which attach trained classifiers to intermediate layers to save inference compute, motivate our second question: LLS already provides such classifiers at no extra cost.

## 3 Implementation Details

### 3.1 Granularity

The reference implementation [2] fixes one head per layer, so we reimplemented it to make the head positions a parameter. A network is described as a list of *units*, each a trainable layer together with its pooling, and a *partition* such as $(2, 5)$ groups consecutive units into blocks. Every block gets one LLS head on its output and its own optimizer, which makes per-layer LLS $(1, \ldots, 1)$ and BP $(L)$ two ends of the same model. During a training step, each block takes the detached output of the previous block, backpropagates its head loss only within itself, and updates; no gradient crosses a block boundary. Peak training memory is therefore set by the largest block, which we measure exactly by counting, with PyTorch's saved-tensor hooks, the bytes that autograd keeps for each block's backward pass.

With this setup we trained all $2^6 = 64$ partitions of VGG8. MobileNetV1 has $2^{13}$ partitions, too many to sweep, so we chose 14 that test the patterns seen on VGG8: uniform blocks, cuts at pooling layers, and merges at the start or end of the network. We also checked seed variance on VGG8, repeated the key partitions on CIFAR100, and tested whether short 10-epoch runs are enough to choose a partition. All runs use schedule-free AdamW [13].

### 3.2 Early Exit with LLS Heads

For each trained network we record the logits of every head on the validation and test sets once, so that stopping rules can be compared offline without rerunning the model. A sample exits at the first head where

the rule fires; its cost is the backbone MACs up to that head plus the cost of every head evaluated. We implemented the standard rules (confidence, entropy, margin and patience) and a *sync* rule of our own, which exits only when the last $w$ heads agree on the class distribution:

$$\frac{2}{w(w-1)} \sum_{k-w<i<j\le k} \cos(p_i, p_j) \;\ge\; \tau. \tag{2}$$

The intuition follows from how LLS trains: every layer is pushed towards the same class-specific pattern, so once consecutive heads give the same distribution, the representation has settled and deeper layers are unlikely to change the decision. Unlike confidence, which trusts a single head, sync requires agreement across depth, so one overconfident head cannot trigger an exit; and unlike patience, which only compares the top class, it compares full distributions, so the heads must also agree on how uncertain they are. For each rule we select, on validation, the cheapest threshold whose accuracy stays within $\delta = 1\%$ of the full network, and report that setting on test. To judge whether fixed heads are good exits, we compare them with linear exits trained on the same frozen features, and with BP networks whose exits are trained either after the backbone (frozen) or together with it (joint).

## 4 Evaluation

### 4.1 Granularity

We compare each partition with two baselines: per-layer LLS, which has the lowest memory, and BP, which has the highest accuracy (Table 1). On VGG8, $(2, 5)$ improves on per-layer LLS by 5.4 points (91.6% vs 86.2%) at 1.29× its memory, and is only 0.7 points below BP while using 40% less memory than BP. On MobileNetV1, $(4, 2, 1 \times 8)$ improves on per-layer LLS by 15.2 points (83.0% vs 67.8%) and is 1.0 point below BP with 69% less memory. Both winners merge the first layers; merging the last four layers of MobileNetV1 is 1.1 points worse than per-layer LLS. We attribute this to the per-layer heads forcing early layers to separate classes before they have built useful features. Partitions with equal memory differ by 2–3 points on VGG8 (seed std 0.1–0.3), and uniform or pooling-aligned blocks are 2–2.5 points below the best, so the partition has to be searched per model; ranking partitions after 10 epochs finds the best one in 4 of 5 memory budgets.

Table 1: Granularity on CIFAR10. $G$: share of the LLS–BP accuracy gap closed; Mem.: peak training memory relative to per-layer LLS.

| VGG8 (60 epochs) | | | | MobileNetV1 (300 epochs) | | | |
|---|---|---|---|---|---|---|---|
| Partition | Acc. | $G$ | Mem. | Partition | Acc. | $G$ | Mem. |
| LLS $1 \times 7$ | 86.2 | 0% | 1.00 | LLS $1 \times 14$ | 67.8 | 0% | 1.00 |
| Uniform $(2, 2, 2, 1)$ | 89.4 | 53% | 1.29 | Merge last 4 | 66.7 | −7% | 2.37 |
| $(1, 1, 5)$ | 89.9 | 60% | 1.00 | Uniform $2 \times 7$ | 77.3 | 58% | 1.54 |
| $(2, 5)$ | 91.6 | 89% | 1.29 | $(4, 2, 1 \times 8)$ | 83.0 | 94% | 2.56 |
| BP | 92.3 | 100% | 2.15 | BP | 84.0 | 100% | 8.21 |

### 4.2 Early Exit

We compare the LLS exits with running the full network, with learned exits, and with BP (Table 2). Against the full network, the LLS exits cut inference MACs by 85–90% on MobileNetV1 and by 32–43% on VGG8 at a cost of at most one point of accuracy. VGG8 saves less because 42% of its MACs are in full-resolution early convolutions that come before its first useful exit. Against per-layer LLS, the coarser partition $(4, 2, 6, 2)$ gives much better exits on MobileNetV1: 82.3% at 10% of the MACs, versus 67.2% at 12%. Against linear exits trained on the same features, the fixed heads are within 0.1 points on both networks without training any exit parameters. Against BP, jointly trained exits are 3.3 points more accurate on MobileNetV1 at similar cost and 2.6 points more accurate on VGG8 at 0.52 instead of 0.68 of the MACs, mainly because the BP networks are more accurate to begin with. Among stopping rules, a

simple margin threshold matches our sync rule on most partitions, since the heads are already calibrated (ECE $\leq 0.08$).

Table 2: Early exit at $\delta = 1\%$ on CIFAR10. MACs: fraction of the full network. LLS partitions: $(4, 2, 6, 2)$ for MobileNetV1, $(3, 3, 1)$ for VGG8.

| | MobileNetV1 | | VGG8 | |
|---|---|---|---|---|
| Exits | Acc. (%) | MACs | Acc. (%) | MACs |
| Fixed LLS heads (ours) | 82.3 | 0.10 | 89.2 | 0.68 |
| Linear probes on LLS | 82.2 | 0.08 | 89.3 | 0.67 |
| BP, frozen exits | 82.8 | 0.10 | 92.1 | 0.70 |
| BP, jointly trained | 85.6 | 0.09 | 91.8 | 0.52 |

### 4.3 Conclusions

A partition determines both the training memory and the exits available at inference, and merging the early layers improves both. The LLS heads are also interpretable in a practical sense: since they are calibrated, their confidence is a reliable signal for exiting i.e., their per-layer accuracy shows where the network stops improving, which tells us where to merge layers and the depth at which a sample exits follows its difficulty [12]. LLS exits do not seem to outperform BP with trained exits, so the case for LLS remains its lower training memory. Since the best partition is model-specific, a natural next step is to search it automatically, as SEDONA [9] does for locally trained blocks.

### References


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